

Lab Goal : This lab was designed to teach you how to use a set.

Lab Description : Read in 2 Sets of numbers. Then, you are to perform the set operations of union, intersection, difference, and symmetric difference on the 2 Sets.

1	2	3	4	5
4	5	6	7	8

UNION – all of the items in both Sets - **1 2 3 4 5 6 7 8**

INTERSECTION – the items that occur in both Sets - **4 5**

DIFFERENCE A-B – the items that occur in A, but not in B - **1 2 3**

DIFFERENCE B-A – the items that occur in B, but not in A - **6 7 8**

SYMMETRIC DIFFERENCE – the items that occur in either Set, but not in both - **1 2 3 6 7 8**

Sample Data :

```
1 2 3 4 5
4 5 6 7 8
10 11 12 13 14 15 16 17
11 13 15 17 19 21 23
4 5 5 6 7 6 7 7 8 8 8 8 8
23 3 4 3 5 3 53 5 46 46 4 6 5 3 4
```

Sample Output :

```
Set one [1, 2, 3, 4, 5]
Set two [4, 5, 6, 7, 8]
```

```
union - [1, 2, 3, 4, 5, 6, 7, 8]
intersection - [4, 5]
difference A-B - [1, 2, 3]
difference B-A - [6, 7, 8]
symmetric difference [1, 2, 3, 6, 7, 8]
```

```
Set one [10, 11, 12, 13, 14, 15, 16, 17]
Set two [11, 13, 15, 17, 19, 21, 23]
```

```
union - [10, 11, 12, 13, 14, 15, 16, 17, 19, 21, 23]
intersection - [11, 13, 15, 17]
difference A-B - [10, 12, 14, 16]
difference B-A - [19, 21, 23]
symmetric difference [10, 12, 14, 16, 19, 21, 23]
```

```
Set one [4, 5, 6, 7, 8, 76]
Set two [3, 4, 5, 6, 23, 46, 53]
```

```
union - [3, 4, 5, 6, 7, 8, 23, 46, 53, 76]
intersection - [4, 5, 6]
difference A-B - [7, 8, 76]
difference B-A - [3, 23, 46, 53]
symmetric difference [3, 7, 8, 23, 46, 53, 76]
```

Useful Collection methods

- `retainAll( )`
- `addAll( )`
- `removeAll( )`